

Scaling Book — Section 6 Exercises

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LLaMA Case-study problems: Counting params / FLOPs

Q1

$$\begin{aligned}\text{Attn params} &= \underbrace{DNH + DKH + DKH + NHD}_{\text{projs}} + \underbrace{D}_{\text{layernorm}}. \\ \text{MLP} &= 2DF + FD.\end{aligned}$$

$$\begin{aligned}\text{Total} &= 2DV + (2DNH + 2DKH + D + 3FD) \cdot L \\ &= 2 \cdot 8192 \cdot 128256 + (2 \cdot 8192 \cdot 64 \cdot 128 + 2 \cdot 8192 \cdot 8 \cdot 128 + 8192 + 3 \cdot 28672 \cdot 8192) \cdot 80 \\ &= 7 \times 10^{10} = 70\text{B params}.\end{aligned}$$

$$\begin{aligned}\text{Total}_{\text{Attn}} &\approx 12 \cdot 10^9 = 12\text{B params}, \\ \text{Total}_{\text{MLP}} &\approx 56.4 \cdot 10^9 = 56\text{B params}, \\ \text{Total}_{\text{vocab}} &\approx 2\text{B params}.\end{aligned}$$

Note: MLP params account for most of the model's weights so we can ignore all other params when reasoning about memory / FLOPs. (Can use simplified transformer layer from prev section.)

Q2

$$\text{FLOPs/token} = 6 \cdot 70\text{B} = 420 \cdot 10^9 \text{ FLOPs}.$$

If we're compute bound + MFU = 1, in bf16 on 1 TPU:

$$T_{\text{comp}} = \frac{420 \cdot 10^9 \text{ FLOPs}}{4.59 \cdot 10^{14} \text{ FLOPs/s}} \approx 10 \cdot 10^{-4} = 1 \text{ ms}.$$

Q3

$$\text{FLOPs}_{\text{Total}} = 6 \cdot 70\text{B} \cdot 15\text{T} = 6 \cdot 70 \cdot 10^9 \cdot 15 \cdot 10^{12} = 6.3 \cdot 10^{24}.$$

Assuming compute bound w/ MFU = 1, bf16 on a single v5p:

$$T_{\text{comp}} = \frac{6.3 \cdot 10^{24}}{4.59 \cdot 10^{14}} = 1.37 \cdot 10^{10} \text{ s}.$$

Q4

$$T_{\text{comp}} = \frac{6.3 \cdot 10^{24}}{8960 \cdot 4.59 \cdot 10^{14} \cdot 0.4} \approx 3.8 \cdot 10^6 \text{ s.}$$

Q5

$$\begin{aligned} \text{Bytes}_{\text{params}} &= 70\text{B} \cdot 2 = 140 \cdot 10^9 \\ \text{Bytes}_{\text{opt}} &= 2 \cdot 4 \cdot 70\text{B} = 560 \cdot 10^9 \end{aligned}$$

(Assuming each checkpoint is of activations size)

$$\begin{aligned} \text{Bytes}_{\text{ckpt}} &= 2 \cdot B \cdot D \cdot L = 2 \cdot 4 \cdot 10^6 \cdot 8192 \cdot 80 \cdot 4 \\ &= 5.24 \cdot 10^{12} \cdot 4 \approx 20.9 \text{ TB.} \end{aligned}$$

$$\text{Total} = 140 \text{ GB} + 560 \text{ GB} + 20.9 \text{ TB} = 21.6 \text{ TB.}$$

$$\# \text{TPUs}_{\text{min}} = \frac{21.6 \cdot 10^3 \cancel{\text{GB}}}{96 \cancel{\text{GB}}} = 225 \text{ TPUs.}$$

Q6

$$\text{Bytes} / \text{chip} = \frac{21.6 \cdot 10^{12}}{8960} = 2.4 \cdot 10^9 = 2.4 \text{ GB.}$$

Sharding LLaMA 3-70B for training

Q7

With no context / sequence parallel, we can only utilize $4 \cdot 10^6 / 4096 \approx 976$ chips. Pure FSDP is not possible here.

We could technically still do it over 976 chips only but it would be a waste for the rest.

Q8

$$\begin{aligned} \text{BS}_{\text{local}} &= \frac{4 \cdot 10^6}{8960} \approx 446. \\ \frac{C}{3 \cdot W_{\text{ici}}} &= \frac{2550}{3} \approx 833. \end{aligned}$$

$446 < 833$ so we would be comm-bounded. FSDP only is still not a good idea.

Q9

$$\frac{X^2}{M_x M_y F} = \frac{(2550)^2}{2 \cdot 28672} \approx 113.$$

$\frac{B}{N} = 446 > 113$ so w/ FSDP+TP we can be compute-bounded.

$$X_{\text{opt}} = \sqrt{\frac{B}{F} \cdot \frac{M_x}{M_y} \cdot N}.$$

Choose $M_x = 2$, $M_y = 1$:

$$X_{\text{opt}} = \sqrt{\frac{(4 \cdot 10^6)}{28672} \cdot 2 \cdot 8960} \approx 1502.$$

$\Rightarrow$ 2048-way FSDP and 4-way TP.

Worked Problems

1) $N = 4 \cdot 8960 = 35840$

When scaling up to multiple pods, we would use pure dp across the pods (we want minimal comm over dcn) and a different parallelism scheme within each pod depending on the new per-pod batch size.

$$\text{BS}_{\text{local}} = \frac{4 \cdot 10^6}{4} = 1 \cdot 10^6.$$

w/ pure TP in each pod:

$$\frac{F}{8960} = \frac{28672}{8960} < \frac{2550}{3} \quad (\text{comm-bounded}).$$

TP + FSDP:

$$\frac{B_{\text{local}}}{N} = 111 < \frac{2550^2}{2F} = 113.$$

Between 4-pods, within each pod we are now comm bounded regardless of what parallelism scheme we choose.

In the intra-pod level, setting $M_x = 2$, $M_y = 1$:

$$X_{\text{opt}} = \sqrt{\frac{B_{\text{local}}}{F} \cdot \frac{M_x}{M_y} \cdot N} = \underbrace{\frac{1}{2} \cdot 1502}_{\substack{\text{b/c our batchsize decreases} \\ \text{by } \frac{1}{4} \text{ which we pull out of the sqrt}}} = 751.$$

$\Rightarrow$ Rounding to the nearest power of 2 gives 512-way FSDP and 16-way TP. (We have chips left over though.)

At the inter-pod level:

$$T_{\text{comm}} = \frac{2 \cdot 2 \cdot 2 \cdot DF}{M \cdot W_{\text{dcn}}}, \quad T_{\text{math}} = \frac{4 \cdot 2BDF}{N \cdot C}.$$

$$T_{\text{math}} > T_{\text{comm}} \Rightarrow \frac{8BDF}{NC} > \frac{8DF}{MW_{\text{dcn}}} \Rightarrow \frac{B \cdot M}{N} > \frac{C}{W_{\text{dcn}}}.$$

$$\frac{4 \cdot 10^6}{4} > 71,360.$$

We will be compute-bounded over dcn.

Assuming MFU = 1, looking at inter-node level and assuming perfect overlap of comp and comm intra-node. Also ignoring that we technically don't use all chips in above parallelism.

$$T_{\text{train}} = \frac{6 \cdot 70B \cdot 15T}{4 \cdot 8960 \cdot 4.59 \cdot 10^{14}} = \frac{6 \cdot 70 \cdot 10^9 \cdot 15 \cdot 10^{12}}{4 \cdot 8960 \cdot 4.59 \cdot 10^{14}} \approx 3.8 \cdot 10^5 \text{ s}.$$

2) Larger model

a) i) Hyperparam table:

hyperparam	value
n_layers (L)	126
d_model (D)	16384
d_{ff} (F)	53248
n_heads (N)	128
n_kv_heads (K)	8
d_qkv (H)	$D/N = 16384/128 = 128$
n_embeddings (V)	128256

ii) Per layer:

$$\text{Attn} = DNH + DKH + DKH + NHD + D = 2DNH + 2DKH + D \approx 2DNH + 2DKH.$$

$$\text{MLP} = 2DF + FD = 3DF.$$

$$\begin{aligned} \text{Total} &= (2DNH + 2DKH + 3DF) \cdot L + 2DV \\ &= (2 \cdot 16384 \cdot 128 \cdot 128 + 2 \cdot 16384 \cdot 8 \cdot 128) \cdot 126 \\ &\quad + 3 \cdot 16384 \cdot 53248 + 2 \cdot 16384 \cdot 128256 \\ &= 4.05 \cdot 10^{11} = 405 \cdot 10^9 \text{ params.} \end{aligned}$$

iii)

$$\text{FLOPs/Token} = 6 \cdot 405 \cdot 10^9 = 2.43 \cdot 10^{12}.$$

iv)

$$\text{FLOPs}_{\text{Total}} = 2.43 \cdot 10^{12} \cdot 15 \cdot 10^{12} = 3.645 \cdot 10^{25}.$$

b) Assuming the actual sequence lengths are $< 8D$ so that attn costs are negligible and that our batch size is 4 mil.

Across the pods I would use pure DP (minimize comm times over dcn):

$$\frac{B \cdot M}{N} = \frac{4 \cdot 10^6}{8} = 5 \cdot 10^5 > \frac{C}{W_{\text{dcn}}} = 73440 \Rightarrow \text{compute bound over dcn.}$$

At the intra-pod level: Pure-DP will not work since model params alone are $405 \cdot 10^9 \cdot 2 = 810$ GB $\gg$ 96 GB HBM / v5p.

Trying pure FSDP:

$$\frac{B_{\text{local}}}{8960} = \frac{(4 \cdot 10^6/8)}{8960} \approx 55 < \frac{2550}{3} = 850.$$

We can't use pure FSDP since we'll be comm-bound.

FSDP+TP (Let $M_x = 2$, $M_y = 1$):

$$\frac{B_{\text{local}}}{N} = \frac{(4 \cdot 10^6/8)}{8960} \approx 55.$$

$$\frac{X^2}{M_x M_y F} = \frac{2550^2}{2 \cdot 53248} \approx 61.$$

We will still be slightly comm-bounded.

$$X_{\text{opt}} = \sqrt{\frac{B_{\text{local}}}{F} \cdot \frac{M_x}{M_y} \cdot N} = \sqrt{\frac{(4 \cdot 10^6/8)}{53248} \cdot 2 \cdot 8960} \approx 410.$$

Closest power of 2 (for wraparound ici) gives us: 448-way FSDP and 20-way TP.

Assuming MFU = 1:

$$T_{\text{train}} = \frac{6 \cdot 405 \cdot 10^9 \cdot 15 \cdot 10^{12}}{8 \cdot 8960 \cdot 4.59 \cdot 10^{14}} \approx 1 \cdot 10^6 \text{ s} \approx 12 \text{ days.}$$